

aws

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United States (N. Virginia)

Account ID: 4969-9996-3738
AWSLabUser-t93eXn7uES6UbM4LfrV47/91226fc...

Amazon Data Firehose

Firehose streams

Create Firehose stream

Info

► Amazon Data Firehose: How it works

Choose source and destination

Specify the source and the destination for your Firehose stream. You cannot change the source and destination of your Firehose stream once it has been created.

Source | Info

Direct PUT

Destination | Info

Amazon S3

Firehose stream name

Firehose stream name

SI-Firehose

Acceptable characters are uppercase and lowercase letters, numbers, underscores, hyphens, and periods.

Transform and convert records - optional

Configure Amazon Data Firehose to transform and convert your record data.

Transform source records with AWS Lambda | Info

Amazon Data Firehose can invoke an AWS Lambda function to transform, filter, decompress, convert and process your source data records. The specified AWS Lambda function can also be used to provide data partitioning keys for the incoming source data before its delivery to the specified destination.

☐ Turn on data transformation

Convert record format | Info

Data in Apache Parquet or Apache ORC format is typically more efficient to query than JSON. Amazon Data Firehose can convert your JSON-formatted source records using a schema from a table defined in [AWS Glue](#). For records that aren't in JSON format, create a Lambda function that converts them to JSON in the Transform source records with AWS Lambda section above.

☒ Enable record format conversion

Output format

☒ Apache Parquet

☐ Apache ORC

Schema for source records

Amazon Data Firehose references table definitions stored in AWS Glue. Choose an AWS Glue table to specify a schema for your source records.

AWS Glue region

United States (N. Virginia)

AWS Glue database

conversion

Use the same S3 bucket and prefix

You can analyze the data delivered by this Firehose stream by querying the AWS Glue table you specified for record format conversion. To ensure this, it is recommended that you choose the same S3 bucket and prefix for your S3 destination that you specified in your AWS Glue table configuration:

S3 bucket: consumption-bucket-496999963738-3f35d650

Prefix: station-data/

AWS Glue table

conversion_table

Version

Latest

Browse

Create table

Decompress source records from Amazon CloudWatch Logs - new | Info

When this feature is turned on, Amazon Data Firehose decompresses Amazon CloudWatch Logs and delivers them to the destination. There will be additional fees for this functionality. For more information, see [Amazon Data Firehose pricing](#). After enabling decompression, you have the option to turn on message extraction to deliver only the content inside the message fields. [Learn more](#).

☐ Turn on decompression

Destination settings

Info

Specify the destination settings for your Firehose stream.

S3 bucket

s3://consumption-bucket-496999963738-3f35d650

Browse

Create

Format: s3://bucket

New line delimiter

You can configure your Firehose stream to add a new line delimiter between records in objects that are delivered to Amazon S3.

☒ Not enabled

☐ Enabled

Dynamic partitioning | Info

Dynamic partitioning enables you to create targeted data sets by partitioning streaming S3 data based on partitioning keys. You can partition your source data with inline parsing and/or the specified AWS Lambda function. You can enable dynamic partitioning only when you create a new Firehose stream. You cannot enable dynamic partitioning for an existing Firehose stream. Enabling dynamic partitioning incurs additional costs per GiB of partitioned data. For more information, see [Amazon Data Firehose pricing](#).

☐ Not enabled

☒ Enabled

You are choosing to enable dynamic partitioning for this Firehose stream. Once this Firehose stream is created, you will not be able to disable dynamic partitioning for this Firehose stream. Dynamic partitioning can result in additional costs.

Multi record deaggregation

Data deaggregation is the process of parsing through the records in a Firehose stream and separating the records based either on valid JSON or on the specified delimiter. If your data is aggregated, dynamic partitioning can be applied only after data deaggregation is performed. So if you enable dynamic partitioning to your aggregated data, you must enable multi record deaggregation.

☒ Not enabled

☐ Enabled

Inline parsing for JSON

To partition source data with inline parsing, you must specify data record parameters to be used as dynamic partitioning keys.

- ☐ Not enabled
- ☒ Enabled

When inline parsing is enabled, you must specify valid JQ expressions and use the specified dynamic partitioning keys in the Amazon S3 bucket prefix in order to create a successful Firehose stream.

Example record

Use the example below to define the dynamic partitioning keys and key values.

Dynamic partitioning keys

Use the fields below to specify key names and JQ expressions to be used as dynamic partitioning keys. Amazon Data Firehose only supports expressions in JQ 1.6.

Key name	JQ expression	
station_id	.station_id	<button>Remove</button>
year	<input %y")"="" type="text" value=".event_timestamp strftime("/>	<button>Remove</button>
month	<input %m")"="" type="text" value=".event_timestamp strftime("/>	<button>Remove</button>
day	<input %d")"="" type="text" value=".event_timestamp strftime("/>	<button>Remove</button>
hour	<input %h")"="" type="text" value=".event_timestamp strftime("/>	<button>Remove</button>
minute	<input %m")"="" type="text" value=".event_timestamp strftime("/>	<button>Remove</button>

Add dynamic partitioning key

Dynamic partitioning keys must be unique. You can add 44 more dynamic partitioning key values.

S3 bucket prefix

For dynamic partitioning, you must use the following expression format in your S3 bucket prefix: `!{namespace:value}`, where namespace can be either `partitionKeyFromQuery` or `partitionKeyFromLambda`, or both. If you are using inline parsing to create the partitioning keys for your source data, you must specify an S3 bucket prefix value that consists of expressions specified in the following format: `!{partitionKeyFromQuery:keyID}`. If you are using an AWS Lambda function to create partitioning keys for your source data, you must specify an S3 bucket prefix value that consists of expressions specified in the following format: `!{partitionKeyFromLambda:keyID}`.

station-data/station_id=!{partitionKeyFromQuery:station_id}/!{partitionKeyFromQuery:year}/!{partitionKeyFromQuery:month}/!{partitionKeyFromQuery:day}/!{partitionKeyFromQuery:hour}/!{partitionKeyFromQuery:minute}/

Apply dynamic partitioning keys

You can repeat the same keys in your S3 bucket prefix. Maximum S3 bucket prefix characters: 1024.

S3 bucket error output prefix

You can specify an S3 bucket error output prefix to be used in error conditions. This prefix can include expressions for Amazon Data Firehose to evaluate at runtime.

error

S3 bucket and S3 error output prefix time zone | Info

Choose a time zone that you want to use for date and time in S3 prefixes

UTC

Retry duration

Specify the retry duration to send partitioned data to S3.

60 seconds

Minimum: 0 seconds, maximum: 7200 seconds. Recommended: 300 seconds.

Buffer hints, compression, file extension and encryption

The fields below are pre-populated with the recommended default values for S3. Pricing may vary depending on storage and request costs.

S3 buffer hints

Amazon Data Firehose buffers incoming records before delivering them to your S3 bucket. Record delivery is triggered once the value of either of the specified buffering hints is reached.

For Firehose streams with dynamic partitioning enabled, we recommend a buffer size of 128 MiB for optimized processing of data. For more information, see [Amazon Data Firehose pricing](#).

Buffer size

The higher buffer size may be lower in cost with higher latency. The lower buffer size will be faster in delivery with higher cost and less latency.

128 MiB

Minimum: 64 MiB, maximum: 128 MiB. Recommended: 128 MiB.

Buffer interval

The higher interval allows more time to collect data and the size of data may be bigger. The lower interval sends the data more frequently and may be more advantageous when looking at shorter cycles of data activity.

60 seconds

Minimum: 0 seconds, maximum: 900 seconds. Recommended: 300 seconds.

Data format conversion is enabled and by default, the Snappy compression method is set to be applied to your data. You cannot enable more than one data compression method at the same time. Use AWS CLI commands to choose another compression method or to disable data compression. [Info](#)

Compression for data records

Amazon Data Firehose can compress records before delivering them to your S3 bucket.

Not enabled

File extension format - optional

Enter the full extension format. If you want to add 'gzip file extension' for your 'json' files, enter '.json.gz'

Enter a file extension (for example, .json, .zip, .json.gz)

File extension must start with a period and may only contain allowed characters: 0-9a-z!-_.~(). File extension cannot exceed 128 characters.

If you specify a file extension, it will override the default file extension that Amazon Data Firehose adds when data format conversion or compression is enabled.

Encryption for data records

Record gets encrypted in the S3 bucket using an AWS Key Management Service (KMS) key.

Use the encryption setting of the S3 bucket

Default encryption for S3 bucket is server-side encryption with Amazon S3 managed keys (SSE-S3). See [S3: Protecting data with server-side encryption](#) for other encryption options.

- ☐ Use SSE-KMS
Server-side encryption with AWS Key Management Service (KMS) keys.

Backup settings

Enabling source data backup ensures that the data can be recovered if record processing transformation does not produce the desired results.

Source record backup in Amazon S3

- ☒ Not enabled
☐ Enabled

▼ Advanced settings

Server-side encryption not enabled; error logging enabled; IAM role KinesisFirehoseRole-3f35d650; no tags.

Server-side encryption [Info](#)

You can use AWS Key Management Service (KMS) to create and manage keys and to control the use of encryption across a wide range of AWS services in your applications.

- ☐ Enable server-side encryption for source records in Firehose stream

Amazon CloudWatch error logging [Info](#)

Choose Enabled if you want Amazon Data Firehose to log record delivery errors to CloudWatch Logs.

- ☐ Not enabled
☒ Enabled

Service access [Info](#)

Amazon Data Firehose uses this IAM role for all the permissions that the Firehose stream needs. To specify different roles for the different permissions, use the API or the CLI.

- ☐ Create or update IAM role **KinesisFirehoseServiceRole-SI-Firehose-us-east-1-1762316508953**
Creates a new role or updates an existing one and adds the required policy to it, and enables Amazon Data Firehose to assume it.
- ☒ Choose existing IAM role
The role that you choose must have policies that include the permissions that Amazon Data Firehose needs.

❗ If you choose an existing IAM role for Amazon Data Firehose, make sure the chosen IAM role has a proper trust policy and permissions required for your source and destination. For more information, see [Controlling Access with Amazon Data Firehose](#).

Existing IAM roles

Only IAM roles with the [required trust policy](#) are available for selection.

KinesisFirehoseRole-3f35d650 

View [KinesisFirehoseRole-3f35d650](#) in the IAM console.

Tags [Info](#)

You can add tags to organize your AWS resources, track costs, and control access.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#)

[Create Firehose stream](#)